Contract and quality management

Sicheng Zhan

Department of Civil Engineering
szhan@hku.hk

Topics to cover this week

1. Contract Fundamentals
2. Payment and variations
3. Time and Risk Allocation
4. Quality Management
5. Safety and integration



What changes after tender award?

Before award:

- Client defines what is required
- Tenderers propose price, approach, and commitments
- Client evaluates competing offers

After award:

- Agreed requirements and commitments become contractual obligations
- Client and contractor must deliver according to the agreed documents
- Changes, payment, time, and acceptance now follow contract procedures

Tendering selects the offer. The contract governs its delivery.



Contract functions

Establishes both the agreed basis for project delivery and the procedures for administering events that arise during execution.

Contract function	Example questions to answer
Define obligations	What must the contractor deliver?
Allocate responsibilities	Who designs? Who provides site information?
Set commercial rules	How and when is the contractor paid?
Provide procedures	What happens if work changes or is delayed?

Make responsibilities and procedures clear before problems occur

- e.g.: an undocumented pipe blocks the proposed route; 10 additional sensors needed; specified equipment is unavailable; work finishes two weeks late



Parties, roles, and authority

During retrofit work, the contractor is asked to stop noisy work because a lecture is taking place. The planned work is now delayed, **what happens next?**

Campus user: stop disruptive work when necessary →
Contractor: stop as required, record the event and notify client →
Designated person: coordinate constraints and arrange resumption →
Contract administration: determine consequences based on the contract

Contractual relationships provide a clear pathway for communication, coordination, and decisions when planned delivery is disrupted.

For group project (scope/requirements/interfaces/contract/risk): recognize operational constraints, consistently reflect on how they should be coordinated, recorded, and resolved



What forms a contract?

The contract is not just a signed agreement, but a coordinated set of documents:

- Contract agreement & conditions
- Scope of works
- Specifications & drawings
- Pricing documents
- Accepted contractor submissions

Look at your tender package, which sections would become part of the contract after award?



Incorporate winning tender

Tender requirement: equipment efficiency $\geq$ 90%
Bidder A: 95% - HK$2.2M
Bidder B: 90% - HK$2.0M
What should be in the contract if A is awarded?

Tender received $\rightarrow$ evaluate $\rightarrow$ clarify deviations/ambiguities $\rightarrow$ identify accepted commitments $\rightarrow$ incorporate relevant commitments $\rightarrow$ execute contract

Not everything should automatically become binding, e.g. completion.
What about a lower-than-requirement cooling system nominal capacity?



How do we pay?

A tender with an estimated cost of HK$2.0M was selected, and construction has been completed successfully. *How much should the client pay?*

A. Exactly HK$2.0M
B. The value of the actual quantities completed
C. The contractor's actual cost + an agreed fee
D. It depends on what the parties agreed

What if the work was changed after award?



Forming a pricing basis

Water pipe replacement with an expected construction cost of around HK$2.0M.

- **Contract A**: complete the defined works for HK$2.0M
- **Contract B**: concrete: HK$850/m³, pipework: HK$420/m, etc., with final quantities to be measured
- **Contract C**: actual allowable cost + 10% fee

A contract does not always fix the final amount. *It defines not only how payment is calculated, but also what is allowed to remain uncertain after contract award*.



Typical arrangements

	Lump Sum	**Remeasurement/ Unit Price**	**Cost Reimbursement**
What is agreed?	Total price for defined work	Unit rates + rules	Reimbursable costs + fee/charge
Final payment basis	Agreed lump sum + adjustment	Actual measured quantities × agreed rates	Actual allowable costs + fee/charge
Suitable when	Scope reasonably defined	Work known, quantities uncertain	Scope/extent highly uncertain
Cost certainty	High	Low	Lowest

How should your project be priced, given how certain your scope and quantities are?



Same project, different basis

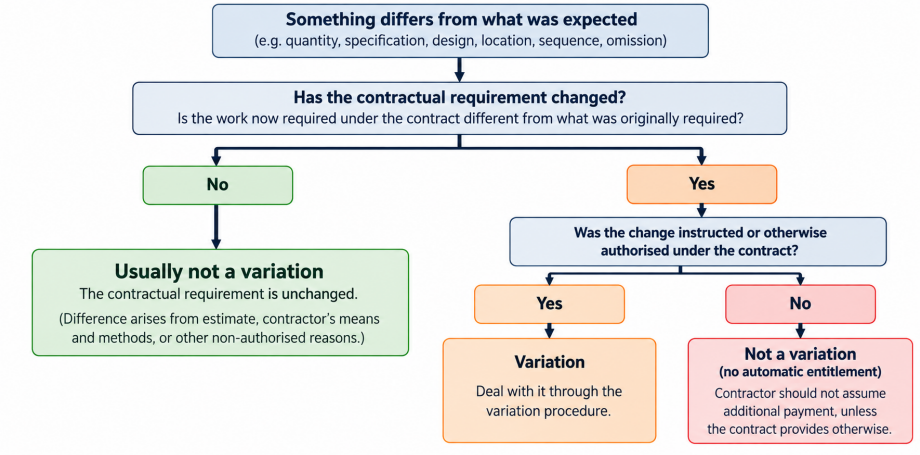
Scope: Repaint all corridor walls on Levels 2–5
Tender-stage estimate: 4000 m²
Actual quantity required to complete the same defined scope: 4300 m²

	Lump Sum	**Remeasurement/ Unit Price**
Tender	HK$400K for the defined works	Estimated 4000 m² × HK$100/m²
Actual quantity	4300 m²	4300 m²
Amount for original scope	HK$400k	HK$430k
Uncertainty mainly born by	Contractor	Client

When can a lump-sum contract price change?

What is a variation?

An authorized change to the work required under the contract



Is this a variation?

BOQ estimates 100 m pipe, but the actual unchanged route measures 115 m under remeasurement.

Contractor underestimated the quantity needed to complete the lump-sum scope.

Contractor installs 15 m extra because it chose a longer route for convenience.

Client instructs the pipe route to extend to another room: +15 m.

Client changes the sensor specification to a high-resolution model.



From variation to valuation

Original contract: 100 sensors @ HK$1000 each
Client instructs 20 additional identical sensors.
How should the additional work be priced?

Existing contract rate applicable → Use where appropriate
Similar work but different conditions → Existing rate may provide a basis for adjustment
No suitable existing rate → Establish a new rate/quotation according to the contract
Work cannot reasonably be priced beforehand → Cost records/daywork may be used after permitted

BOQ and SOR are not merely prepared to compare tenders.



When does the contractor get paid?

Contract award → Mobilization → Work progresses → Work progresses → Completion → Final account

Payment	What it is	Typical basis
Advance payment	Payment before corresponding work has been completed	Agreed upfront amount, often linked to mobilization
Interim payment	Periodic payment during construction	Value of completed work, with contractual adjustments
Final payment	Settlement after completion/final account	Settles the remaining contractual entitlement

Does your project require large upfront procurement or mobilization that could justify an advance payment?



Interim payment

Value of work properly completed V.S. the contractor's monthly expenditure

Contractor performs work → Measures/values work completed to date → Submits payment application → Contract Administrator assesses/certifies → Client pays certified amount

Example: **value of work completed to date**: HK$800k
+ **valued variations**: HK$50k
- **retention**: HK$40k
- **previous payments**: HK$500k
= **current payment**: HK$310k

Retention: part of the amount temporarily withheld as security for proper completion and defect correction



Final account

Original pricing basis → Quantity remeasured → Authorized variations → Other contractual adjustments → Final contractual entitlement

Example: **Original tender sum**: HK$2.00M
+ **Remeasurement adjustment**: HK$0.04M
+ **Valued variation**: HK$0.08M
- **Omitted work**: HK$0.04M
= **Final amount**: HK$2.08M

A variation increases the price, but what if it also delays completion?



Program delay v.s. contractual delay

Does contractor-side delay automatically mean a contract violation?

Project commencement: **1 May** ; Planned completion: **15 June**; Contractual completion: **30 June**

Event → affected activity → program impact → impact on completion

Being behind program is a project management problem. Missing the contractual completion date is a contractual problem.

What if completion is delayed for reasons beyond the contractor's responsibility?



Causes for delay

Change in work → Cost consequence + Time consequence

What else could prevent the contractor from completing on time?

Source	Example
Contractor-side	Poor planning, labor shortage, late subcontractor, equipment ordered too late, etc.
Client-side	Late site access, late information/decision, instructed additional work, etc.
External/neutral	Unexpected weather, authority restrictions, other events outside either party's control

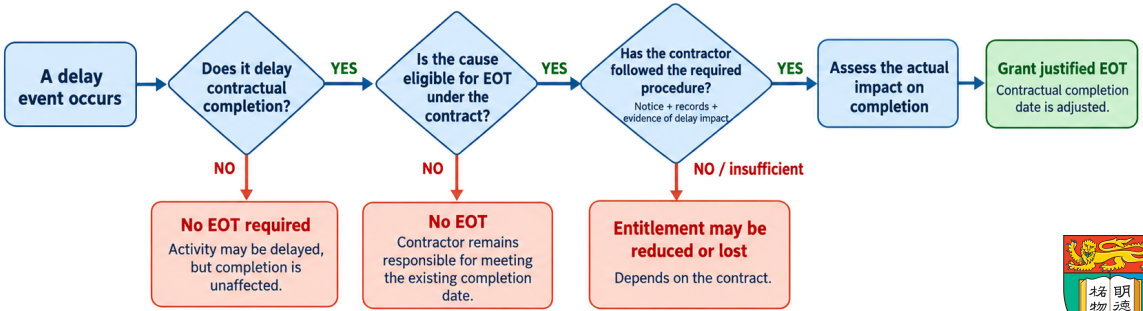
If completion is delayed, should the contractor still be required to meet the original completion date?

Different cause → different contractual consequences



Extension of Time

An adjustment to the contractual completion date when a qualifying event delays completion



Is an EOT required?

Contractor's subcontractor arrives 7 days late.

Client gives site access 5 days later than agreed.

Client instructs additional work that delays commissioning.

Heavy rain delays external works.

Painting is delayed 4 days but completion remains on time.



More time ≠ More money

If the contractor is granted an EOT, does the client also need to pay more?

Time entitlement: should the contractual completion date be extended?
Cost entitlement: should the contractor be compensated for additional cost caused by the delay?

Delay event	Additional time?	Compensation?
Contractor's own planning	Usually No	Usually No
Agreed site access delayed	Potentially Yes	Potentially Yes
Client instructs additional work	Potentially Yes	Potentially Yes
Qualified extreme weather	Potentially Yes	Not necessarily

So after establishing EOT, who bears the cost consequence of the delay?



When the contractor finishes late

Original completion date: 30 Sept
Approved EOT: +10 days
Actual completion: 18 Oct
How many days of contractor delay are against the contractual completion obligation?

Delay event → justified EOT assessed → completion date adjusted → actual completion compared with adjusted date → remaining contractor-responsible delay may trigger contractual consequences

Liquidated damages: pre-agreed damages for late completion, where provided by the contract



Risk allocation with contract

Responsibility → Mitigation → Contractual consequence

Late subcontractor mobilization → **Mitigation**: resource planning and early procurement → **Allocation**: contractor → **Time**: no EOT → **Cost**: contractor bears additional damage/cost

Previously agreed work area becomes unavailable → **Mitigation**: early coordination with campus users → **Allocation**: client-side if access was contractually promised → **Time**: potential EOT → **Cost**: potential compensation, subject to the contract

From "who owns the risk?" to "what happens if it actually occurs?"



Same event, different contractual consequence

During retrofit work, the contractor is asked to stop noisy work because a lecture is taking place. Completion is delayed.

The restriction was clearly stated in the tender documents

- Contractor should have planned around it
- Contractor bears potential cost/damage

HKU unexpectedly suspends previously permitted work

- Client-side event
- Potential EOT and possibly compensation, depending on the contract

The same physical event can produce different contractual consequences depending on the agreed responsibilities, information, and procedures.



What good quality means

If the contractor installs exactly as specified, is the work automatically "good quality"?

- **Conformance**: meets drawings, specifications, and contractual requirements
- **Performance**: actually achieves the intended outcome
- **Workmanship/finish**: installed to an acceptable standard
- **Fitness for use**: works properly in the intended operating context

Quality is judged against defined acceptance criteria, not appearance or intention alone.
E.g., a ventilation system is installed according to drawings, but measured airflow is below the required level.



Quality starts before construction

Specify → Plan → Approve → Inspect → Test → Correct → Accept

- **Quality planning**: what standards, checks, and responsibilities are required?
- **Quality assurance**: are the right systems and procedures in place?
- **Quality control**: does the actual work comply with the requirements?

E.g., lighting retrofit: Specification (minimum illuminance/efficiency) → Submittal (proposed luminaire model) → Inspection (installation and wiring) → Test (lighting level/controls) → Acceptance (results meet requirements)

Inspection at the end cannot compensate for unclear requirements at the beginning.



Inspection and Test Plan

ITP: a structured plan stating what will be checked, when, by whom, against what criteria, and what evidence is required.

Stage	Before installation	Before concealment
Check	Equipment submittal	Services inspection
Acceptance criterion	Meets specified performance	No defects/correct routing
Responsibility	Contractor/client	Client rep/inspector
Evidence	Approved submittal	Hold-point record

Hold point - Work cannot proceed until inspection/approval is completed.
Witness point - Client may attend, but work may proceed according to the agreed procedure if attendance is waived.



When the work does not comply

Installed sensor model does not meet the specified accuracy.
What happens next?

Non-conformance identified

↓

Record/report

↓

Assess significance

↓

Correct/replace/approved concession if permitted

↓

Reinspect/retest

↓

Close out

Is this a variation?



Commissioning

Installed $\neq$ operational $\neq$ accepted

Inspection

Is it installed correctly?

Testing

Does an individual component/system meet specified requirements?

Commissioning

Does the complete system operate correctly under intended conditions?

Demonstrate that the completed system performs as a system, not just that individual components have been installed correctly.

Finishing the work ≠ handover

Physical completion

- Works complete
- Defects/non-conformance addressed

Technical evidence

- Test / commissioning records
- Certificates

Operational information

- As-built drawings
- O&M manuals
- Warranties
- Training where required

Handover transfers a completed asset together with the information needed to operate and maintain it.



From hazard to control

Hazard identified → Risk assessed → Control measures defined → Method statement/permit/access arrangement → Work performed → Monitor and revise if conditions change

Examples:

- Work at height
- Hot work
- Electrical isolation
- Dust/noise near occupied rooms
- Temporary access obstruction
- Lifting equipment in public areas

What are the 2–3 most important safety interfaces in your retrofit, and what controls should the contractor plan for?



Who is responsible?

Safety responsibility is distributed across the project, but the contractor normally controls day-to-day site execution.

Client

Provide relevant site information and constraints; coordinate access and interfaces; appoint competent parties and establish project requirements

Contractor

Plan and execute work safely; assess hazards and implement controls; supervise workers and subcontractors; maintain safe site conditions

Workers/subcontractors

Follow procedures and controls; report hazards and incidents

Safety ↔ time, cost, and quality

Safety is not a separate checklist, but interacts directly with program, cost, quality, and operations.

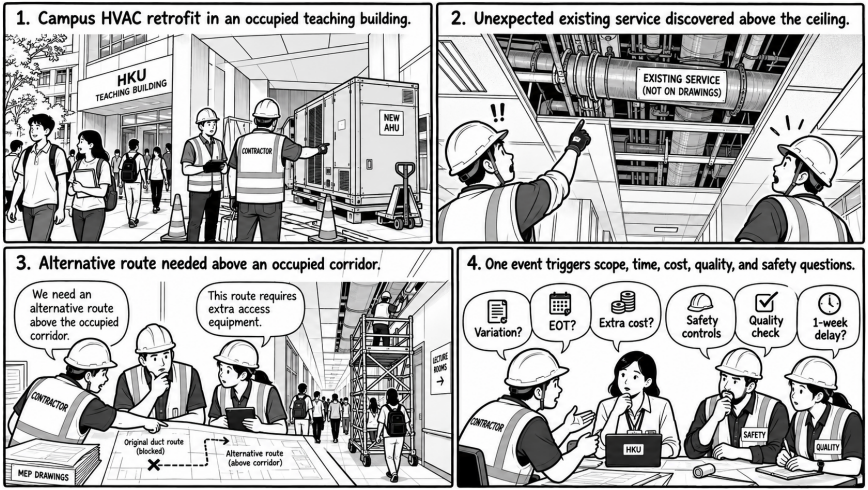
Late safety approval → work cannot start → possible program impact

Unsafe access route → temporary works / revised sequence required → possible cost and delay impact

Rushed installation to recover delay → increased safety and quality risk



Integrated case: HVAC retrofit



Questions to be answered

Scope - Is rerouting already within the contractor's original obligation?
Scope - If not, is a variation required?
Time - Does the event delay contractual completion?
Time - Is EOT potentially available?
Cost - Who bears the additional labor, access equipment, and prolongation cost?
Cost - Is compensation available?
Quality - Does the revised route still meet technical/performance requirements?
Quality - What inspection or testing is needed?
Program - How will work above the occupied corridor be controlled?
Program - Does access need to be restricted?
Program - Who coordinates with building users?



Final project checks

Scope: Is it clear what the contractor must deliver?

Procurement: Is the selected route and tendering method justified?

Pricing: Is the pricing basis appropriate and usable?

Contract: Are roles, procedures, changes and delays addressed?

Quality: Are requirements measurable and verifiable?

Safety & operations: Are key campus constraints and controls identified?

Handover: Is it clear what evidence is required before acceptance?

